Yajvan Ravan

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EDUCATION

Massachusetts Institute of Technology (MIT) *June 2025*

B.Sc. in Electrical Engineering & Computer Science; ***GPA: 5.0/5***

- **Selected Courses:** Design & Analysis of Algorithms. Machine Learning. Computer Vision. Inference. Representation, Reasoning, & Inference in AI. Matrix Methods. Real Analysis. Linear Algebra. Sensorimotor Learning

EXPERIENCE

FortyFive Labs–*Technical Staff* *Jun 2025 - Dec 2025*

- MIT CSAIL spin-off building open-source robotics tool suite and data hub
- Deployed Redis-based task queue into AWS and workstations allowing image generation at 50,000 images/hour
- Set up 20-node (5090 + A6000) in-house cluster with SLURM/Ansible for robot policy training/testing

Isola Lab, MIT–*Research Intern (Advisor: Phillip Isola)* *Oct 2024 - May 2025*

- Built LucidXR, a VR data engine (>100k lines) to scale learning-from-demonstration for manipulation tasks
- Compiled MuJoCo sim to WebAssembly for in-browser physics simulation; developed visualization in ThreeJS.
- Implemented generative model policies (diffusion, VAE, transformer) with automated sim eval and video logging
- Trained visuomotor policies across 8 manipulation tasks (incl. deformable material, dexterous hands).
- Designed MuJoCo environment schema and XML tooling for rapid scene creation and validation.
- Developed data pipeline converting raw MoCap poses to 10D action representations (Gram-Schmidt matrices).
- Developed keypoint trajectory augmentation (10x data amplification) and 3D Gaussian Splat evaluation suite
- Built distributed synthetic imagery generation pipeline with StableDiffusion + ControlNet + ComfyUI.
- **Achieved zero-shot sim-to-real transfer (zero-shot generalization) on pick-and-place.**
- **Co-first author publication in CoRL 2025**. https://lucidxr.github.io/

LIS Group, MIT–*Robotics Research (Advisors: Leslie Kaelbling, Tomás Lozano-Pérez)* *Aug 2023 - Sep 2024*

- Developed PoPi, a hierarchical system for long-horizon mobile manipulation with unknown dynamics.
- Built data collection/processing system for Boston Dynamics Spot: synchronized RGB-D streams and proprioception, AprilTag-based localization, robot-to-world frame transforms
- Trained state-based, conditional UNet diffusion policy for short-horizon control within 2m.
- Implemented A*/RRT motion planners for SE(2) and offline mapping (clustering for obstacle detection).
- Designed hierarchical planner-policy architecture and full deployment stack; achieved 80% success on furniture rearrangement (vs. 0% pure learning, 50% pure planning). https://yravan.github.io/plannerorderedpolicy/.

NASA Langley Research Center–*Research Intern* *Jun 2023 - Dec 2023*

- Built real-time wildfire detection system on multispectral satellite imagery (IR + thermal). Processed raw sensor data and trained UNet segmentation models. **First-author paper, AIAA SciTech 2026.**

AWARDS

Gold Medal at the 2019 International Chemistry Olympiad – **1st** in the US, **19th** in the world
2020 USA Math Olympiad Qualifier and **2019 USA Junior Math Olympiad Qualifier** – **Top 500** in US

PUBLICATIONS

Ravan, Y.*, Rashid, A.*, et al. Lucid-XR: An Extended-Reality Data Engine for Robotic Manipulation. *Conference on Robot Learning (CoRL) 2025.* https://lucidxr.github.io/.

Ravan, Y., et al. Combining Planning and Diffusion for Mobility with Unknown Dynamics. arXiv Preprint arXiv:2410.06911. https://yravan.github.io/plannerorderedpolicy/.

Yu, A.*, Yang, G.*, ... , **Ravan, Y.**, ... LucidSim: Learning Visual Parkour from Generated Images. *Conference on Robot Learning (CoRL) 2024.* https://lucidsim.github.io/.

Ravan, Y., et al. Real-Time Wildfire Localization on the NASA AMS using Deep Learning. *AIAA SciTech 2026.*

PROJECTS

MITScript Interpreter (C++): Interpreter with garbage collection, bytecode compiler, custom VM. Page
BeaverNav (Python): 10K+ line navigation app with pathfinding on 300+ floor plans at MIT. Page

SKILLS

Languages: C++, Python, C, JavaScript **Robotics:** Motion planning (A*, RRT, SE(2)), MuJoCo, Drake, IsaacLab **Systems:** Linux, Git, Docker, SSH, Web Assembly, SLURM **ML/Perception:** PyTorch, TensorFlow, OpenCV, diffusion models, Weights and Biases.